



魅力统计

Fascinating Statistics

第五章 统计基础与思想

Chapter 5 Statistical Basics and Ideas

Chapter 5 Statistical Basics and Ideas

- 5.1 Statistics
- 5.2 Parameter Estimation
- 5.3 Hypothesis Testing



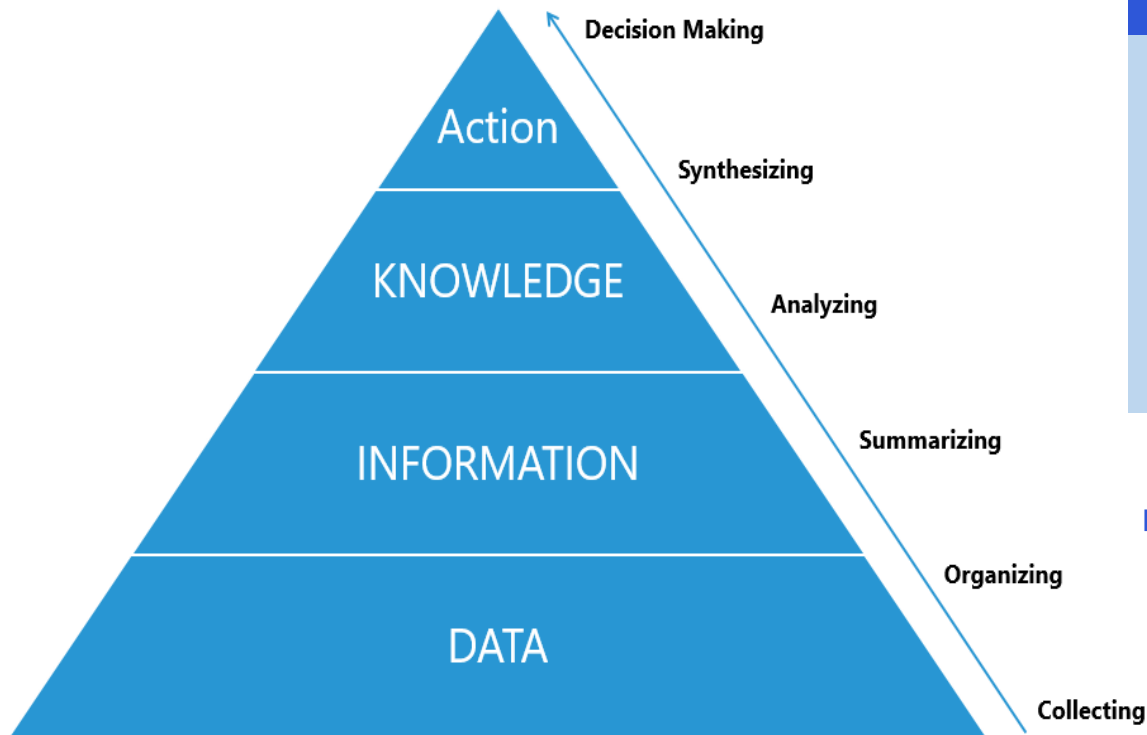
5.1 Statistics

- By the end of Chapter 3, we mentioned that statistical inference is the process of using data to infer the probability model that generated the data.
- In Chapter 4, we introduced concepts of population and sample, where a sample is represented by r.v. $X_1, X_2, \dots, X_n$ before observation and values $x_1, x_2, \dots, x_n$ afterwards.
- Although the sample contains information about the population, this information is not given directly and needs to be summarized.
 - E.g., randomly select 50 male students in SUSTech and let $X_1, X_2, \dots, X_{50}$ be their weights, how to evaluate the overall weight of male students at SUSTech through this sample?
 - Naturally, you would consider doing some calculations, e.g., $(X_1 + \dots + X_{50})/50$, $\max\{X_1, \dots, X_{50}\}$, $\min\{X_1, \dots, X_{50}\}$, etc.
 - These summaries based on samples are called **statistics (统计量)**.



5.1 Statistics

- To extract information from the sample (i.e., data), we need to process the data, i.e., calculate $g(X_1, \dots, X_n)$.



Statistic

- If $X_1, \dots, X_n$ is a sample from population $X \sim f(x; \theta)$ and $g(x_1, \dots, x_n)$ is an n -variate function (n 元函数). Define random variable

$$T = g(X_1, \dots, X_n),$$

- then T is called a **statistic (统计量)** if $g(\cdot)$ does not involve any unknown parameter.

- Here are some commonly used statistics:

- **Sample mean (样本均值)**

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$



5.1 Statistics

- Commonly used statistics:

- Sample variance (样本方差) and sample standard deviation (样本标准差)

Why not divide by n ? $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad S = \sqrt{S^2}.$

- Order statistics (顺序统计量)

$$(X_{(1)}, X_{(2)}, \dots, X_{(n)}).$$

$$X_{(1)} = \min\{X_1, \dots, X_n\}, \\ X_{(n)} = \max\{X_1, \dots, X_n\}$$

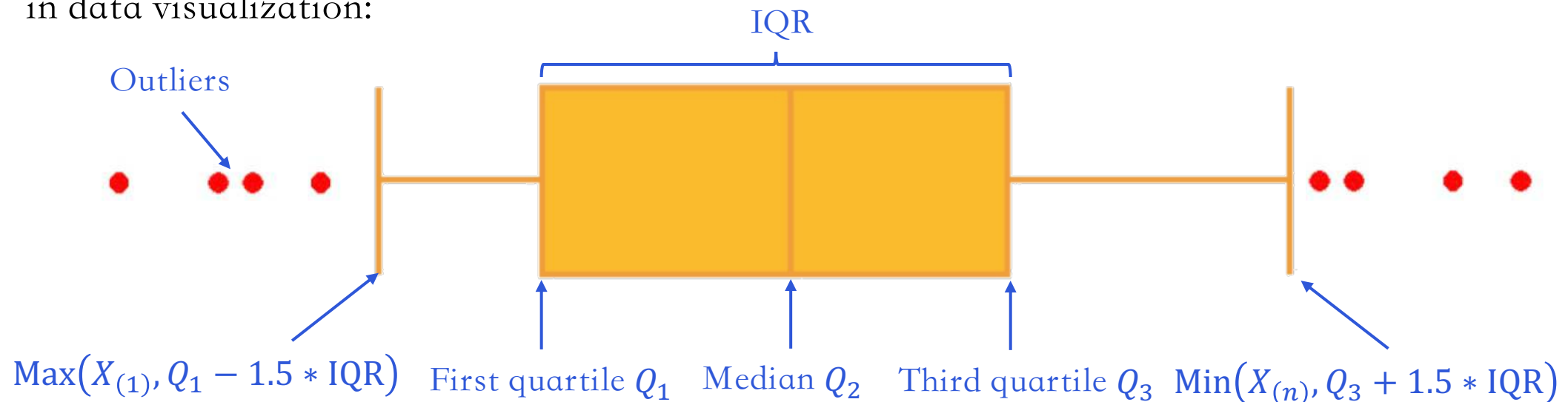
- Sample p -quantile (样本 p 分位数), number that exceeds at most $100p\%$ of the sample and is exceeded by at most $100(1-p)\%$ of the sample

$$Q_p = \begin{cases} X_{(\lfloor np+1 \rfloor)}, & \text{if } np \text{ is not an integer} \\ \frac{(X_{(np)} + X_{(np+1)})}{2}, & \text{if } np \text{ is an integer} \end{cases},$$



5.1 Statistics

- Specifically:
 - $Q_{0.5}$ is the sample median (样本中位数), also denoted as Q_2 .
 - $Q_{0.25}$ is the sample lower/first quartile (样本下/第一四分位数), also denoted as Q_1 .
 - $Q_{0.75}$ is the sample upper/third quartile (样本上/第三四分位数), also denoted as Q_3 .
 - $Q_3 - Q_1$ is the sample interquartile range (样本四分位距, sample IQR).
- A typical application of order statistics and sample quantiles is the box plot (箱线图) learned in data visualization:



5.1 Statistics

- Different statistics have their own characteristics and applicable scenarios.
 - E.g., as descriptions of the population's central position, the sample mean is easily affected by outliers because it involves each individual in the sample, while the sample median is relatively robust and insensitive to outliers.

2023年居民收入和消费支出情况

2024/01/17 10:00 来源：国家统计局



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一、居民收入情况

2023年，全国居民人均可支配收入39218元，比上年名义增长6.3%，扣除价格因素，实际增长6.1%。分城乡看，城镇居民人均可支配收入51821元，增长（以下如无特别说明，均为同比名义增长）5.1%，扣除价格因素，实际增长4.8%；农村居民人均可支配收入21691元，增长7.7%，扣除价格因素，实际增长7.6%。

按收入来源分，2023年，全国居民人均工资性收入22053元，增长7.1%，占可支配收入的比重为56.2%；人均经营净收入6542元，增长6.0%，占可支配收入的比重为16.7%；人均财产净收入3362元，增长4.2%，占可支配收入的比重为8.6%；人均转移净收入7261元，增长5.4%，占可支配收入的比重为18.5%。

2023年，全国居民人均可支配收入中位数33036元，增长5.3%，中位数是平均数的84.2%。其中，城镇居民人均可支配收入中位数47122元，增长4.4%，中位数是平均数的90.9%；农村居民人均可支配收入中位数18748元，增长5.7%，中位数是平均数的86.4%。



猿大侠 >

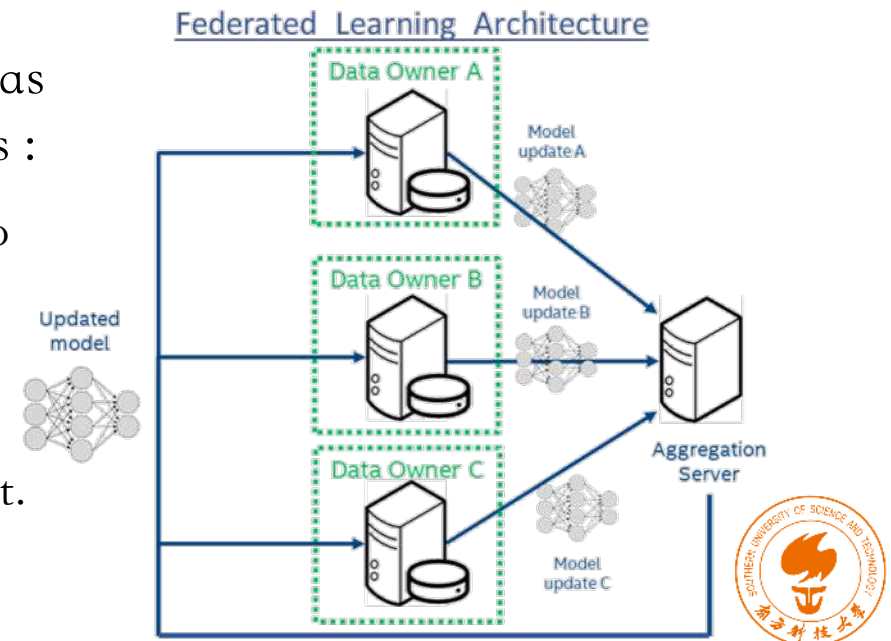


截至2023年6月30日为止的6个月，腾讯薪酬共支出540.69亿元，平均每位员工51.74万元。照此计算，腾讯员工平均年薪可达100万！



5.1 Statistics

- When the form of the population distribution is known but contains unknown parameters, the inference of its parameters is called **parametric statistical inference** (参数统计推断).
- Statistics can be considered as the processing of samples and can be used to infer unknown parameters of the population distribution.
- Then, is there a statistic that can retain all the information about the unknown parameters in the original sample?
- Such a statistic is called a **sufficient statistic** (充分统计量), it has not only theoretical significance, but also practical applications :
 - Sufficient statistics can be used when several data owners want to jointly make statistical inferences but don't want to share the original data.
 - One solution is to only share the values of sufficient statistics to protect data privacy and aggregate them to obtain the final result.
 - This is the basic idea of **distributed computing** (分布式计算) and **federated learning** (联邦学习).



Chapter 5 Statistical Basics and Ideas

- 5.1 Statistics
- 5.2 Parameter Estimation
- 5.3 Hypothesis Testing



5.2 Parameter Estimation

- The first task of parametric statistical inference is to estimate the unknown parameters in the population distribution based on a sample, i.e., **parameter estimation** (参数估计).

Example 5.1

A factory produced a large batch of products. It randomly sampled n products and found k defective products. How to estimate the defective rate of the entire batch of products?

Thinking

- Let r.v. X denote the status of a randomly selected product, assume that it follows a Bernoulli distribution, i.e., **Bernoulli(p)**, then p is the defective rate, which is the parameter to be estimated.
- The status of the n products is denoted as $X_1, X_2, \dots, X_n$, by the LLN, when n is large, the sample mean $\bar{X}$ will approach the population mean p , so $\bar{X}$ can be used as an **estimator** (估计量) of p .
- After observation, the sample observed values $x_1 + x_2 + \dots + x_n = k$, so $\bar{x} = k/n$ is the **estimated value** (估计值) of p .



5.2 Parameter Estimation

- Estimating an unknown parameter with a statistic is called **point estimation (点估计)**.

Point Estimation

- Let $X_1, \dots, X_n$ be a simple random sample from the population $X \sim f(x; \theta_1, \dots, \theta_k)$, $\theta_1, \dots, \theta_k$ are the unknown parameters.
 - If $\hat{\theta}_l(X_1, \dots, X_n)$ is a statistic used to estimate θ_l , then $\hat{\theta}_l$ is called an **estimator (估计量)** of θ_l .
 - Plugging in the sample observed values, $\hat{\theta}_l(x_1, \dots, x_n)$ is called an **estimate** or **estimated value (估计值)** of θ_l .
 - Both $\hat{\theta}_l(X_1, \dots, X_n)$ and $\hat{\theta}_l(x_1, \dots, x_n)$ can be abbreviated as $\hat{\theta}_l$.
- There may be multiple estimators for the same parameter, e.g., for a normal population $N(\mu, \sigma^2)$, the estimators of the population mean μ can be
$$\hat{\mu}_1 = \bar{X}, \quad \hat{\mu}_2 = Q_2, \quad \hat{\mu}_3 = \frac{1}{2}(X_{(1)} + X_{(n)}).$$
 - A natural question is, which estimator is better? We first introduce some properties used to compare multiple point estimators.



5.2 Parameter Estimation

- Since an estimator $\hat{\theta}_l$ is a statistic, i.e., a random variable, its value is sometimes greater and sometimes smaller than the true parameter value θ_l .
- So, a natural criterion to evaluate whether $\hat{\theta}_l$ is a good estimator is to see if it underestimate or overestimate θ_l on average.

Unbiasedness (simplified version)

- Let $X_1, \dots, X_n$ be a simple random sample from the population $X \sim f(x; \theta_1, \dots, \theta_k)$ and $\hat{\theta}_l(X_1, \dots, X_n)$ is an estimator of θ_l . If

$$E(\hat{\theta}_l) = \theta_l,$$

- then $\hat{\theta}_l$ is called an **unbiased estimator** (无偏估计量) of θ_l . Otherwise, it is called a **biased estimator** (有偏估计量) of θ_l .
- $E(\hat{\theta}_l) - \theta_l$ is called the **bias** (偏差) of the estimator.



5.2 Parameter Estimation

Example 5.2

- Let $X_1, X_2, \dots, X_n$ be a simple random sample from population X , the population mean and variance are $E(X) = \theta_1$ and $\text{Var}(X) = \theta_2$ (θ_1, θ_2 are unknown parameters).
- Show that the sample mean $\bar{X}$ and sample variance S^2 are unbiased estimators of θ_1 and θ_2 .

Solution

$$E(\bar{X}) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n E(X_i) = \theta_1.$$

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{\theta_2}{n}.$$

$$\therefore (n-1)S^2 = \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n [(X_i - \theta_1) + (\theta_1 - \bar{X})]^2 = \sum_{i=1}^n (X_i - \theta_1)^2 - n(\bar{X} - \theta_1)^2$$

$$\therefore (n-1)E(S^2) = \sum_{i=1}^n E(X_i - \theta_1)^2 - nE(\bar{X} - \theta_1)^2 = \sum_{i=1}^n \theta_2 - n \frac{\theta_2}{n} = (n-1)\theta_2 \quad \therefore E(S^2) = \theta_2.$$



5.2 Parameter Estimation

- When comparing multiple estimators of the same parameter, we prefer those that are unbiased.
- However, unbiased estimator may not be unique.
 - E.g., for the population $X \sim \text{Poisson}(\lambda)$, $\hat{\lambda}_1 = \bar{X}$ and $\hat{\lambda}_2 = S^2$ are both unbiased estimators of λ .
 - Moreover, for any constant c , $c\hat{\lambda}_1 + (1 - c)\hat{\lambda}_2$ is also an unbiased estimator of λ .
- Then a natural question is: how to compare multiple unbiased estimators?
- Unbiasedness suggests that an estimator fluctuates around the true parameter, considering the stability of estimation, we would like the magnitude of fluctuation to be as small as possible.

Relative Efficiency (simplified version)

- Let $X_1, \dots, X_n$ be a simple random sample from the population $X \sim f(x; \theta_1, \dots, \theta_k)$, $\hat{\theta}_{l1}$ and $\hat{\theta}_{l2}$ are two unbiased estimators of θ_l . If

$$\text{Var}(\hat{\theta}_{l1}) \leq \text{Var}(\hat{\theta}_{l2}),$$

- then $\hat{\theta}_{l1}$ is said to be **more efficient (更有效)** than $\hat{\theta}_{l2}$.



5.2 Parameter Estimation

- Then we introduce two common methods for obtaining point estimates:
 - Method of moments (or moment estimation, 矩估计法)
 - Maximum likelihood estimation (MLE, 极大似然估计法)
- A brief introduction of the method of moments:
 - **Moments (矩)** are the expectations of powers of a random variable, $\mu_k = E(X^k)$ is the k th population moment (总体 k 阶矩), $\tilde{\mu}_k = E(X - E(X))^k$ is the k th population central moment (总体 k 阶中心矩).
 - It is called “moment” because it has a similar concept in physics and engineering.
 - We also have sample moments, e.g., k th sample moment (样本 k 阶矩) and k th sample central moment (样本 k 阶中心矩):

$$M_k = \frac{1}{n} \sum_{i=1}^n X_i^k, \tilde{M}_k = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^k.$$

- The method of moments was proposed by **Karl Pearson**, and the idea is to **use sample moments to estimate population moments**, and its theoretical basis is the **Law of Large Numbers**.



5.2 Parameter Estimation

Example 5.3

- Computer chips produced by a factory have a certain type of rare defects. Suppose that the number of defects on a randomly selected chip X follows a Poisson distribution with unknown parameter λ .
- A simple random sample of computer chips is drawn and the number of defects on each chip is recorded, denoted as $X_1, \dots, X_n$. How to estimate the unknown parameter λ based on the sample?

Solution

- Since $\lambda = E(X)$ is the 1st population moment, we can estimate it with the 1st sample moment, i.e., $\bar{X}$.
- On the other hand, $\lambda = \text{Var}(X)$ is also the 2nd population central moment, we can estimate with the 2nd sample central moment, i.e., $\sum_{i=1}^n (X_i - \bar{X})^2 / n = (n-1)S^2 / n$.
- Moreover, consider the 2nd population moment $\mu_2 = E(X^2) = \text{Var}(X) + [E(X)]^2 = \lambda + \lambda^2$, we have:

$$\lambda = \sqrt{\mu_2 + \frac{1}{4} - \frac{1}{2}} \xrightarrow{\text{estimate } \mu_2 \text{ with the 2nd sample moment}} \sqrt{\frac{1}{n} \sum_{i=1}^n X_i^2 + \frac{1}{4} - \frac{1}{2}} \text{ is also a moment estimator of } \lambda.$$



5.2 Parameter Estimation

- The method of moments only uses the moments of a distribution, but not the form of the distribution.
- So, the method of moments may not make full use of all information about the parameter in the population distribution.
- The maximum likelihood estimation (MLE) directly uses the form of the population distribution.
- The idea of MLE is very straightforward: if a model parameter can generate the observed data with the highest likelihood, then this model parameter is the most reasonable one.

Example 5.4

- There are several red balls and white balls in a bucket. Suppose the ratio of the two colors of balls is 1:3, but it is not clear whether there are more red balls or white balls.
 - By drawing three balls from the bucket independently with replacement, we obtain: red, red, white.
 - How to determine the ratio of red balls to white balls in the bucket?
- Actually, the definition of **likelihood function** (似然函数) is used in this example.



5.2 Parameter Estimation

Likelihood Function and Maximum Likelihood Estimator (MLE)

- Let $\mathbf{X} = (X_1, \dots, X_n)$ be a simple random sample from the population $X \sim f(x; \boldsymbol{\theta})$ and the sample observed values are $\mathbf{x} = (x_1, \dots, x_n)$. Θ is the parameter space, i.e., the set of values that $\boldsymbol{\theta}$ can take.
- The **likelihood function** (似然函数) is a **function of $\boldsymbol{\theta}$** , measuring the likelihood of observing $\mathbf{X} = \mathbf{x}$. “Likelihood” means “probability” for the discrete case, and “density” for the continuous case:

$$\mathcal{L}(\boldsymbol{\theta}; x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \boldsymbol{\theta}), \boldsymbol{\theta} = (\theta_1, \dots, \theta_k) \in \Theta.$$

- If there exist $\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}(x_1, \dots, x_n)$ such that

$$\mathcal{L}(\hat{\boldsymbol{\theta}}) = \max_{\boldsymbol{\theta} \in \Theta} \mathcal{L}(\boldsymbol{\theta}; x_1, \dots, x_n),$$

Maximizing $\mathcal{L}(\boldsymbol{\theta}; \mathbf{x})$ is equivalent to maximizing the log of it

- then the corresponding $\hat{\boldsymbol{\theta}}(X_1, \dots, X_n)$ is the **maximum likelihood estimator** (极大似然估计量) of $\boldsymbol{\theta}$.
- In everyday language, “likelihood” and “probability” are often used interchangeably as synonyms for “chance”.
- However, in mathematics, Fisher’s adoption of the term “likelihood” deliberately highlights that his conceptual framework—though deeply intertwined with probability—is distinct.



5.2 Parameter Estimation

Example 5.5

- Suppose that the population X follows the following discrete distribution

Value	0	1	2	3
Probability	θ^2	$2\theta(1 - \theta)$	θ^2	$1 - 2\theta$

- where $0 < \theta < 0.5$ is an unknown parameter.
- Now we have the observed values of a simple random sample from the population: 3, 1, 3, 0, 3, 1, 2, 3.
- Derive the moment estimator and MLE of θ and calculate its estimated value based on the sample.

Solution

- Consider the 1st population moment of X :

$$\mu_1 = E(X) = 0 \times \theta^2 + 1 \times 2\theta(1 - \theta) + 2 \times \theta^2 + 3 \times (1 - 2\theta) = 3 - 4\theta \Rightarrow \hat{\theta} = \frac{3 - M_1}{4} = \frac{3 - \bar{X}}{4}.$$

Moment estimator of θ

- Based on the sample observed values, we have

$$\bar{x} = (3 + 1 + 3 + 0 + 3 + 1 + 2 + 3)/8 = 2 \Rightarrow \text{the estimated value of } \theta \text{ is } \hat{\theta} = \frac{1}{4}.$$



5.2 Parameter Estimation

Solution

- Given the sample observed values, the likelihood is:

$$\mathcal{L}(\theta; \mathbf{x}) = \theta^2 \times [2\theta(1 - \theta)]^2 \times \theta^2 \times [(1 - 2\theta)]^4 = 4\theta^6(1 - \theta)^2(1 - 2\theta)^4.$$

- The log-likelihood function (对数似然函数) is

$$\ell(\theta; \mathbf{x}) = \log \mathcal{L}(\theta; \mathbf{x}) = \log 4 + 6 \log \theta + 2 \log(1 - \theta) + 4 \log(1 - 2\theta).$$

- Take the first derivative of $\ell(\theta; \mathbf{x})$ and set it to zero:

$$0 = \frac{d\ell(\theta; \mathbf{x})}{d\theta} = \frac{6}{\theta} - \frac{2}{1 - \theta} - \frac{8}{1 - 2\theta} = \frac{24\theta^2 - 28\theta + 6}{\theta(1 - \theta)(1 - 2\theta)}.$$

- Solve the equation $24\theta^2 - 28\theta + 6 = 0$, we obtain

$$\hat{\theta} = \frac{7 \pm \sqrt{13}}{12}.$$

- Since $0 < \theta < 0.5$, the maximum likelihood estimated value of θ is

$$\hat{\theta} = \frac{7 - \sqrt{13}}{12}.$$



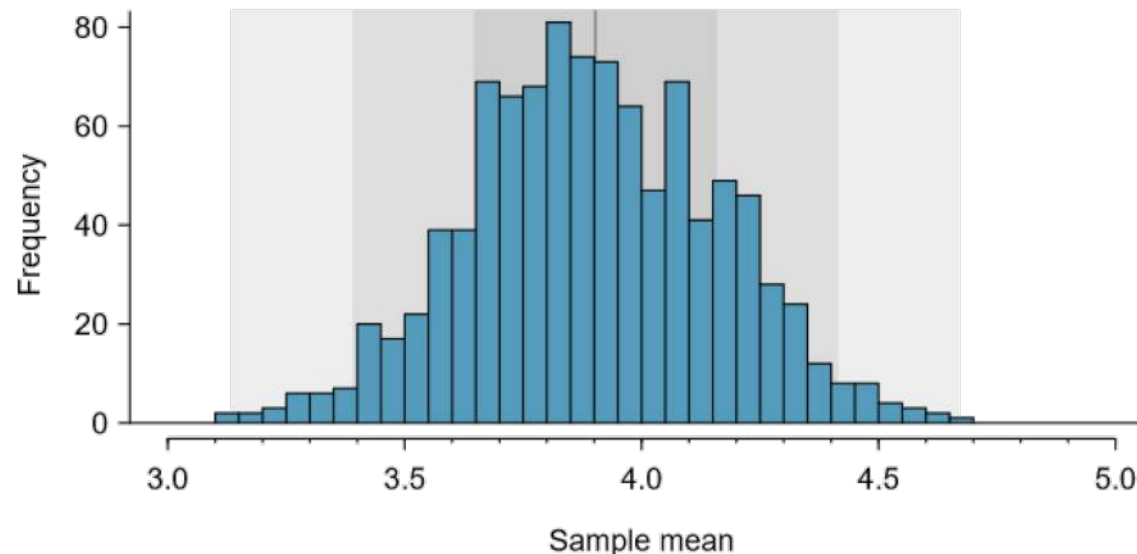
5.2 Parameter Estimation

- Keep in mind that a point estimator $\hat{\theta}(X_1, X_2, \dots, X_n)$ is a random variable, which also follows some probability distribution.
 - E.g., if $X_1, X_2, \dots, X_n$ is a simple random sample from the population $X \sim N(\mu, \sigma^2)$, then
 - E.g., if $X_1, X_2, \dots, X_n$ is a simple random sample from the population $X \sim \text{Bernoulli}(p)$, then

$$\hat{\mu} = \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

$$\hat{p} = \bar{X} \sim_{\text{approx.}} N\left(p, \frac{p(1-p)}{n}\right).$$

By the Central Limit Theorem, when n is large



5.2 Parameter Estimation

- Given the sample observed values, a point estimate provides a concrete estimated value of the parameter. However, the accuracy of this estimate is not provided by the point estimate itself.
- Interval estimation is proposed to address this limitation.
- Estimating the range of a quantity of interest is very common in daily life, e.g.:
 - During an investigation, the height of a suspect might be estimated to be between 175cm and 180cm.
 - The return rate of a financial product might be estimated to fall between 2.6% and 3.2%.



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5.2 Parameter Estimation

- Naturally, we may want to estimate the range of a population parameter based on a sample.
- The general idea of **interval estimation (区间估计)** is
 - Find two statistics $\hat{\theta}_1(X_1, \dots, X_n) \leq \hat{\theta}_2(X_1, \dots, X_n)$ and use the **random interval** $(\hat{\theta}_1, \hat{\theta}_2)$ to estimate the range of θ .
 - Since $(\hat{\theta}_1, \hat{\theta}_2)$ is a random interval while θ is a fixed value, $(\hat{\theta}_1, \hat{\theta}_2)$ may not cover the true value of θ .
 - If the width of the interval is large, then it has higher probability of covering the true value of θ , but its precision is relatively low.
 - Therefore, we need to balance between the coverage probability and precision.
- The **confidence interval (CI, 置信区间)** is a widely used interval estimation proposed by **Jerzy Neyman** in 1937.
- The Bayesian school proposes the **credible interval (可信区间)** as another approach to interval estimation.



5.2 Parameter Estimation

- In 1988, an article published in the BMJ (British Medical Journal) indicated that medical journals were required to report confidence intervals.

The NEW ENGLAND JOURNAL of MEDICINE

Estimating with confidence

The *BMJ* now expects scientific papers submitted to it to contain confidence intervals when appropriate.¹ It also wants a reduced emphasis on the presentation of P values from hypothesis testing.² The *Lancet*,^{3,4} the *Medical Journal of Australia*,⁵ and the *American Journal of Public Health*⁶ have implemented the same policy, and it has been endorsed by the International Committee of Medical Journal Editors.⁷ One of the blocks to implementing the policy has been that the methods needed to calculate confidence intervals are not readily available in most statistical textbooks. Today the *BMJ* continues a series of articles that aims at filling that gap (p 1238); they will eventually be published as a book. Further articles in the *American Journal of Public Health* and the *Annals of Internal Medicine* have debated the uses of confidence intervals and hypothesis tests and discussed the interpretation of confidence intervals.⁸⁻¹⁴

ORIGINAL ARTICLE

Ribociclib plus Endocrine Therapy in Early Breast Cancer

BACKGROUND

Ribociclib has been shown to have a significant overall survival benefit in patients with hormone receptor (HR)–positive, human epidermal growth factor receptor 2 (HER2)–negative advanced breast cancer. Whether this benefit in advanced breast cancer extends to early breast cancer is unclear.

RESULTS

As of the data-cutoff date for this prespecified interim analysis (January 11, 2023), a total of 426 patients had had invasive disease, recurrence, or death. A significant invasive disease–free survival benefit was seen with ribociclib plus an NSAI as compared with an NSAI alone. At 3 years, invasive disease–free survival was 90.4% with ribociclib plus an NSAI and 87.1% with an NSAI alone (hazard ratio for invasive disease, recurrence, or death, 0.75; 95% confidence interval, 0.62 to 0.91; $P=0.003$). Secondary end points — distant disease–free survival and recurrence-free survival — also favored ribociclib plus an NSAI. The 3-year regimen of ribociclib at a 400-mg starting dose plus an NSAI was not associated with any new safety signals.



5.2 Parameter Estimation

Confidence Interval (CI)

- Let $X_1, \dots, X_n$ be a simple random sample from the population $X \sim f(x; \theta)$.
 - For $\forall \alpha \in (0, 1)$, if there exists two statistics $\hat{\theta}_1(X_1, \dots, X_n)$ and $\hat{\theta}_2(X_1, \dots, X_n)$ such that
$$P(\hat{\theta}_1 < \theta < \hat{\theta}_2) \geq 1 - \alpha,$$
 - then $(\hat{\theta}_1, \hat{\theta}_2)$ is called a **confidence interval of θ with confidence level $1 - \alpha$** (置信水平为 $1 - \alpha$ 的置信区间), or simply a **$100(1 - \alpha)\%$ confidence interval**.
 - $\hat{\theta}_1$ and $\hat{\theta}_2$ are called the **confidence lower limit** (置信下限) and **confidence upper limit** (置信上限), respectively.
-
- Commonly used values of α are 0.1, 0.05, 0.01, corresponding to 90%, 95%, 99% CIs.
 - Question:** Do you have any idea about how to construct a confidence interval?
 - Let's illustrate with a simple example.



5.2 Parameter Estimation

Example 5.6

- $X_1, \dots, X_n$ is a simple random sample from $X \sim N(\mu, \sigma^2)$, where the value of σ^2 is known. Derive the $100(1 - \alpha)\%$ confidence interval of the unknown parameter μ .

Solution

- A natural idea to construct a confidence interval is to start from a point estimator and form an interval by adding and subtracting a quantity around the point estimate.
- For this example, a good point estimator of μ is $\bar{X}$, it can be obtained by both the method of moments and method of maximum likelihood, it is a consistent and unbiased estimator of μ .
- By the definition of a $100(1 - \alpha)\%$ CI, we would like to determine the value of $c_1 > 0, c_2 > 0$, s.t.,

$$P(\bar{X} - c_1 < \mu < \bar{X} + c_2) = 1 - \alpha,$$

- with an equivalent expression $P(\mu - c_2 < \bar{X} < \mu + c_1) = 1 - \alpha$.
- Obviously, the value of c_1, c_2 need to be determined by referring to the distribution of $\bar{X}$, which is

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$



5.2 Parameter Estimation

Solution

- So, c_1, c_2 can be determined by

$$1 - \alpha = P(\mu - c_2 < \bar{X} < \mu + c_1) = P\left(\frac{-c_2}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{c_1}{\sigma/\sqrt{n}}\right) = P\left(-\frac{\sqrt{n}c_2}{\sigma} < Z < \frac{\sqrt{n}c_1}{\sigma}\right),$$

- where $Z \sim N(0, 1)$. The c_1, c_2 satisfying the equation above and with the minimum width of the interval $c_1 + c_2$ is

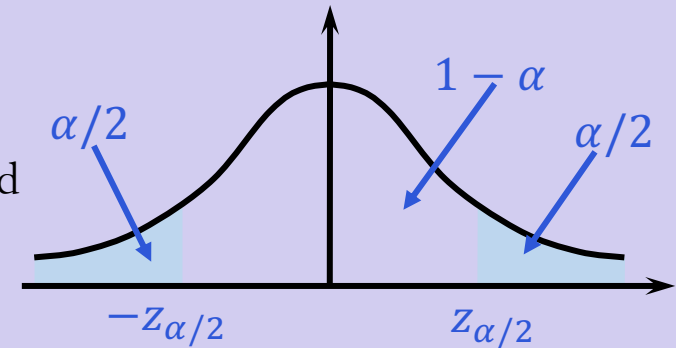
$$c_1 = c_2 = z_{\alpha/2} \frac{\sigma}{\sqrt{n}},$$

- where $z_{\alpha/2}$ is the **upper $\alpha/2$ -quantile (上 $\alpha/2$ 分位点)** of the standard normal distribution, whose value can be found in the $N(0,1)$ table.

- Therefore, the $100(1 - \alpha)\%$ CI of μ (when σ^2 is known) is

$$\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right).$$

- With the value of $z_{\alpha/2}$, σ^2 and observed values $x_1, \dots, x_n$ of the sample, the confidence interval can be calculated.



5.2 Parameter Estimation

Example 5.6 (Continued)

- $X_1, \dots, X_n$ is a simple random sample from $X \sim N(\mu, \sigma^2)$, given $\sigma^2 = 1$ and sample observed values 4.6, 4.2, 5.0, 3.1, 3.4, 2.4, 4.4, 3.2, 3.9, 4.0, calculate the 95% confidence interval of μ .

Solution

- From the standard normal distribution table, we obtain $z_{0.025} = 1.96$.
- Based on the sample observed values, we have $\bar{x} = 3.82$, plugging in $\sigma^2 = 1$, $n = 10$, and $z_{0.025} = 1.96$, the 95% CI of μ is

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \approx 3.82 \pm 0.62 = (3.20, 4.44).$$

- **Question:** How to understand the 95% confidence interval (3.20, 4.44)?

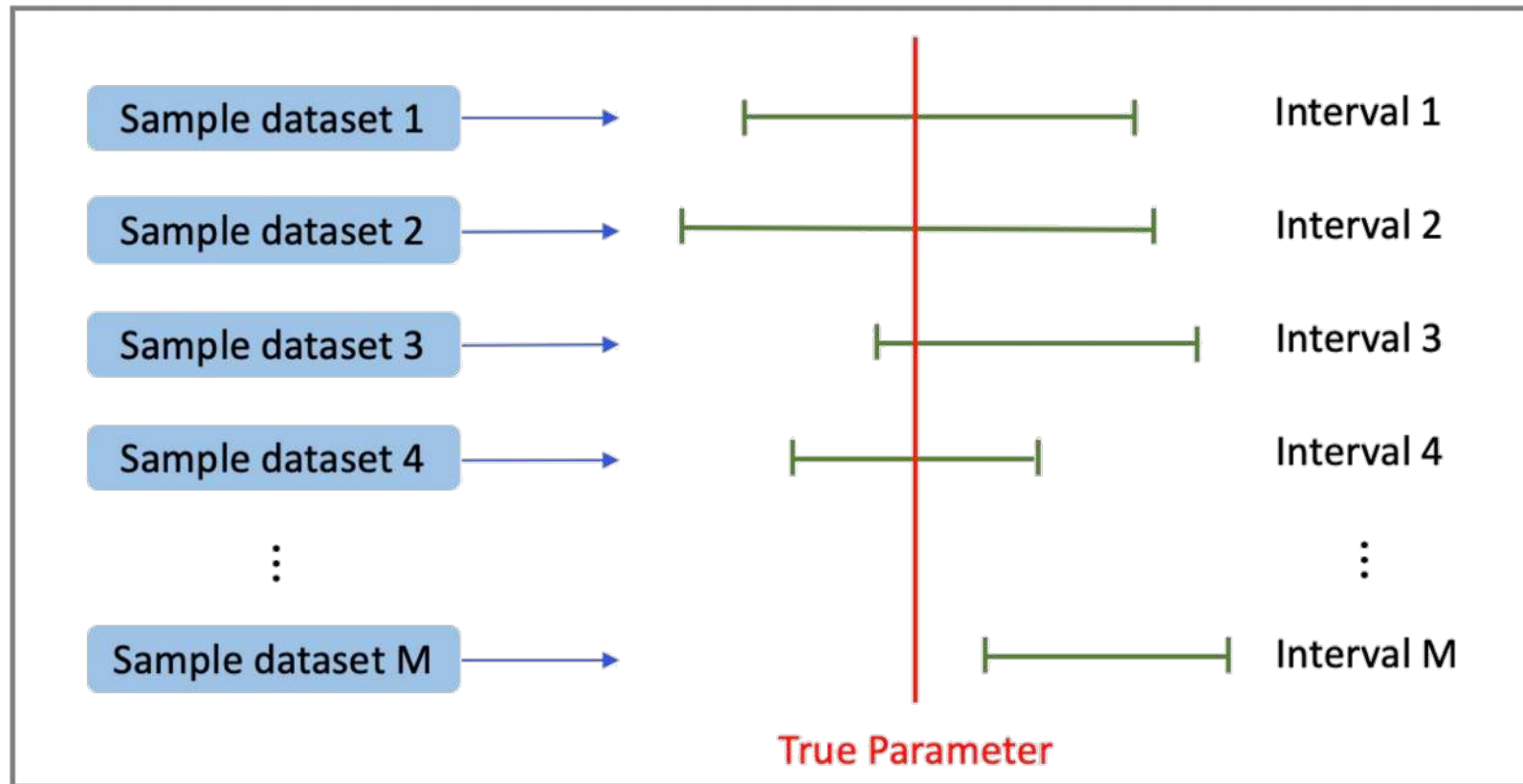
- The probability that (3.20, 4.44) covers the true value of μ is 0.95? 
- 95% of the sample observed value fall into (3.20, 4.44)? 

(3.20, 4.44) either covers or fails to cover the true value of μ



5.2 Parameter Estimation

- 95% confidence interval $(\hat{\theta}_1, \hat{\theta}_2)$ is: in repeated sampling, 95% of all the 95% confidence intervals obtained will cover the true value of the parameter.



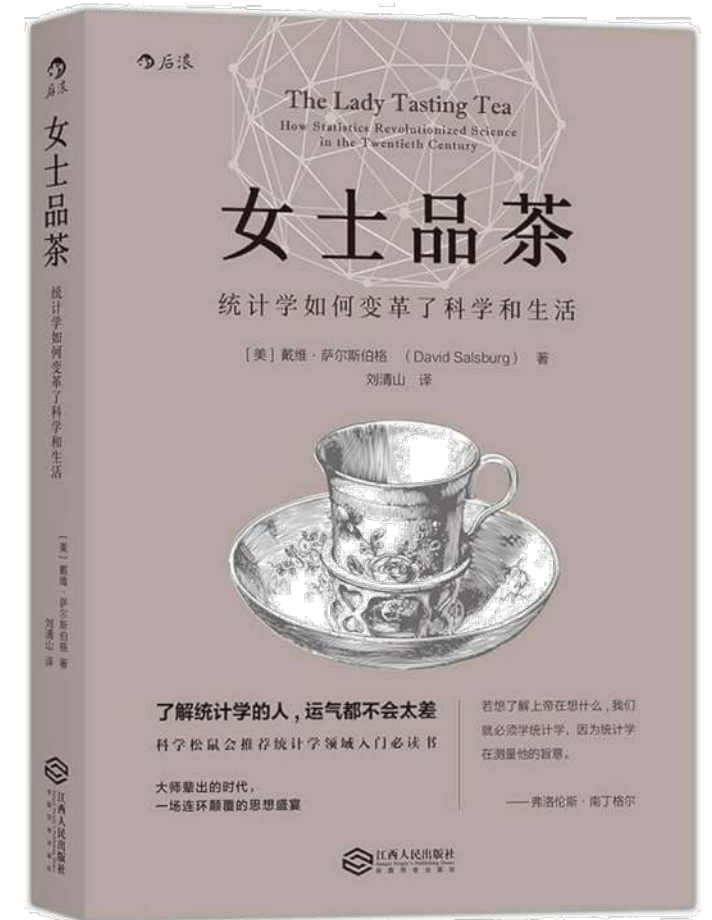
Chapter 5 Statistical Basics and Ideas

- 5.1 Statistics
- 5.2 Parameter Estimation
- 5.3 Hypothesis Testing



5.3 Hypothesis Testing

- Let's start this chapter with a historical story.
 - On a summer afternoon in the late 1920s in Cambridge, England, a group of university faculty members, along with their friends, were gathered around an outdoor table enjoying afternoon tea.
 - A lady insisted that the taste of tea is different depending on whether the milk is added to the tea or the tea added to the milk. However, the scientists present thought this was unlikely.
 - A man named Fisher fell into deep thought and then suggested testing the lady's claim through an experiment.
 - They brewed tea in different ways and asked her to identify how each cup of milk tea had been made.
 - The details of this experiment were recorded in Fisher's 1935 book *The Design of Experiments*.
 - Fisher did not describe the outcome of the experiment, but it is said that the lady correctly identified the method used for each cup of tea.

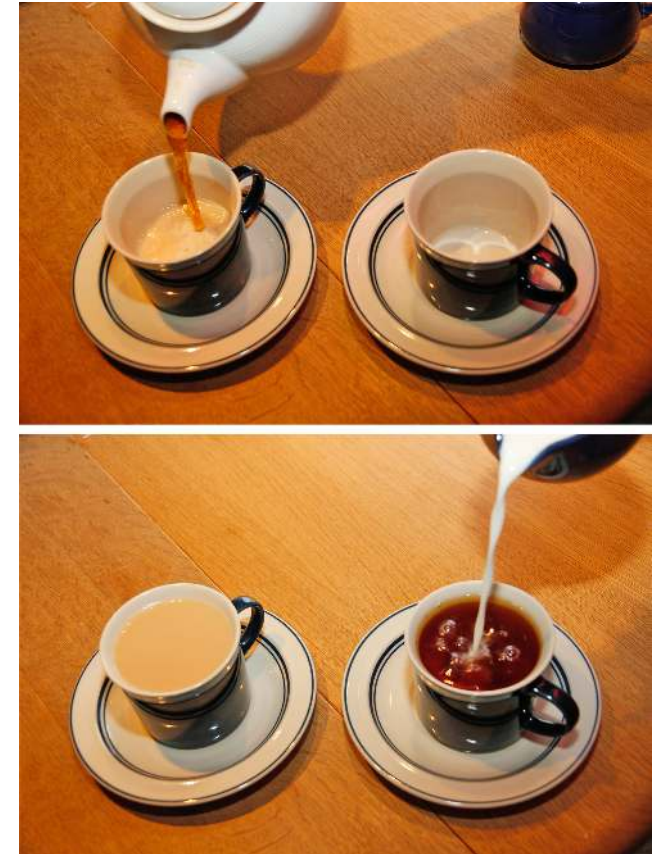


5.3 Hypothesis Testing

- Let's briefly understand the basic ideas of hypothesis testing through the example of the lady tasting tea.

Example 5.7

- There are a total of 8 cups of milk tea: 4 cups where tea is added first, and 4 cups where milk is added first. The cups are identical and are presented to the lady in a random order for tasting.
- After tasting all 8 cups, the lady indicated which ones had the tea added first, and the remaining ones are assumed to have had the milk added first.
- How can we determine from her tasting results whether the lady can truly distinguish between the two methods of making the milk tea?



5.3 Hypothesis Testing

Solution

- Suppose that the lady knew that there are 4 cups with tea added first among the 8 cups of milk tea, let the number of correctly identified cups among those with tea added first by the lady be denoted as X .
- It is easy to see that X can take five possible values: 0, 1, 2, 3, or 4.
- The larger the value of X , the better the lady's ability to distinguish the two ways of making milk tea.
- But at what value of X can we be more confident that the lady can indeed tell the difference between the two ways of making the milk tea?
- Let's assume that the lady has no ability to distinguish the two ways, meaning her guesses are completely random, with equal probabilities of guessing correctly or incorrectly. Under this assumption, the probability of each of the five possible values of X can be calculated.
- First, the number of ways to randomly select 4 out of 8 cups of milk tea is

$$\binom{8}{4} = \frac{8!}{4!4!} = 70,$$

- then we count the number of ways $X = 0, 1, 2, 3, 4$ and calculate the corresponding probabilities.



5.3 Hypothesis Testing

Solution

X	Number of ways	Prob.
0	$\binom{4}{0} \times \binom{4}{4-0} = \frac{4!}{4! \times 0!} \times \frac{4!}{0! \times 4!} = 1$	0.0143
1	$\binom{4}{1} \times \binom{4}{4-1} = \frac{4!}{3! \times 1!} \times \frac{4!}{1! \times 3!} = 16$	0.2286
2	$\binom{4}{2} \times \binom{4}{4-2} = \frac{4!}{2! \times 2!} \times \frac{4!}{2! \times 2!} = 36$	0.5143
3	$\binom{4}{3} \times \binom{4}{4-3} = \frac{4!}{1! \times 3!} \times \frac{4!}{3! \times 1!} = 16$	0.2286
4	$\binom{4}{4} \times \binom{4}{4-4} = \frac{4!}{0! \times 4!} \times \frac{4!}{4! \times 0!} = 1$	0.0143

- It can be seen that under the assumption that the lady has no ability to distinguish between the two ways, the probability of her correctly identifying the 4 cups where tea was added first is only 0.0143.
- This probability is very small, so we are more inclined to believe that she does have the ability to distinguish.
- On the other hand, if she correctly identifies 3 or fewer cups where tea was added first, we would conclude that there is insufficient evidence to prove she has any distinguishing ability.



5.3 Hypothesis Testing

- A vital role of Statistics is in verifying a hypothesis (a statement / claim / conjecture).
- Based on a random sample (i.e., data), we would like to verify whether
 - a lady can really distinguish between the two ways of making milk tea;
 - the average internet speed is 54 Mbps, as claimed by the internet service provider;
 - the proportion of defective products is at most 3%, as promised by the manufacturer;
 - a hardware upgrade was efficient;
 - the supporting rate of a candidate in town A is higher than that in town B.
 - a new medical treatment is more effective than a standard treatment.
- Hypothesis Testing is a statistical inference method used to determine whether data supports a specific hypothesis. Hypothesis testing involves two opposing hypotheses:
 - **Null hypothesis (零假设/原假设)**: typically the default assumption, representing no effect, no difference, or no association, denoted as H_0 . It is tested through data collection to decide whether to reject it or not.
 - **Alternative hypothesis (备择假设)**: usually the researcher's hypothesis, representing the presence of an effect, a difference, or an association, denoted as H_1 .



5.3 Hypothesis Testing

- H_0 and H_1 are simply two mutually exclusive statements, e.g.,
 - H_0 : a suspect is innocence $\leftrightarrow H_1$: a suspect is guilty.
 - H_0 : a new drug is not effective $\leftrightarrow H_1$: a new drug is effective.
 - H_0 : a hardware upgrade is not efficient $\leftrightarrow H_1$: a hardware upgrade is effective.
- In the process of hypothesis testing, we test H_0 based on the sample data collected. If the sample data contradicts with H_0 , then we reject H_0 .
- The question it wants to address is whether the results shown in the data are caused by pure randomness in sampling.



Example 5.8

- A student wanted to know whether the mean height of all male students in SUSTech was greater than 175cm. He randomly selected 25 male students from SUSTech and recorded their height, the average height was 177cm.
- Can we conclude that the mean height of all male students in SUSTech is greater than 175cm?



5.3 Hypothesis Testing

- We will introduce some basic concepts in hypothesis testing through the hypothetical example.
 - Let X be the population of the height of all male students in SUSTech, and assume that $X \sim N(\mu, 5^2)$, then the hypothesis to be tested is:

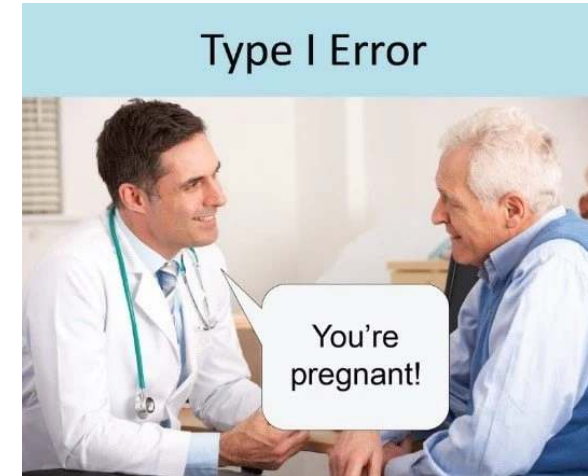
$$H_0: \mu \leq 175 \leftrightarrow H_1: \mu > 175.$$

- Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be a simple random sample from the population X , then a natural decision-making strategy is: reject H_0 when $\bar{X} - 175 > c$, where c is a constant to be determined.
- The set of samples $\mathbf{X} = (X_1, \dots, X_n)$ such that $\bar{X} - 175 > c$, denoted by $\{\mathbf{X}: \bar{X} - 175 > c\}$, is called the **rejection region (RR, 拒绝域)** of the testing problem.
- Since $\bar{X}$ is a random variable, a decision made on whether to reject H_0 based on $\bar{X}$ is a **random decision**, while whether H_0 is true or not is deterministic (though unknown).
- This suggest that our decision might be wrong.
- Therefore, we may make two types of errors in general hypothesis testing problems:
 - **Type I error (第I类错误)**: we reject H_0 when it is true.
 - **Type II error (第II类错误)**: we fail to reject H_0 when it is not true.



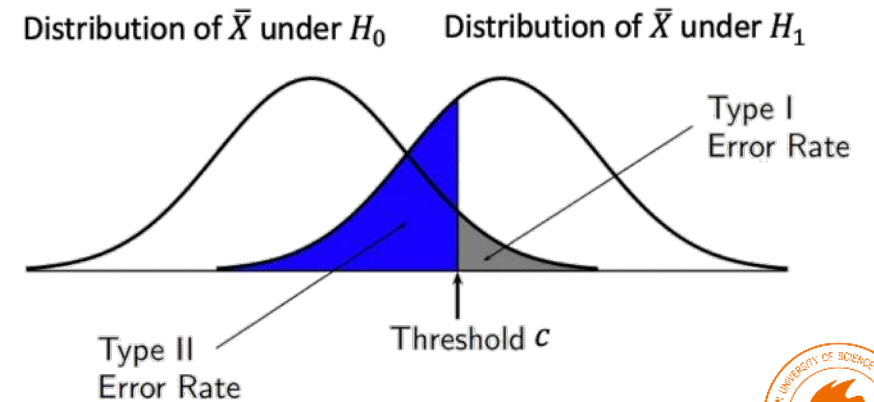
5.3 Hypothesis Testing

	H_0 is true	H_1 is true
Fail to reject H_0	Correct Decision	Type II Error (False Negative, 假阴性)
Reject H_0	Type I Error (False Positive, 假阳性)	Correct Decision



5.3 Hypothesis Testing

- Due to the randomness in sampling, errors cannot be completely avoided. However, we should try to make the probability of errors lower.
 - **Type I error rate (第I类错误率)**: the probability of making a Type I error, $P(X \in RR | H_0 \text{ is true})$.
 - **Type II error rate (第II类错误率)**: the probability of making a Type II error, $P(X \notin RR | H_1 \text{ is true})$.
- It is noted that these two error rates are **traded-off against each other**, i.e., they can not be reduced simultaneously. To be more specific:
 - For a **given sample size** and **form of the RR**, we cannot reduce the two error rates at the same time.
 - E.g., the RR for Example 5.8 is derived to be $\{X: \bar{X} - 175 > c\}$, as the value of c changes, one of the two error rates becomes larger and the other becomes smaller.
 - Therefore, when determining the value of c , we need to **balance between the two error rates**.



5.3 Hypothesis Testing

- In hypothesis testing, we typically try to protect H_0 because it serves as the default or baseline assumption, ensuring that any departure from H_0 is justified by strong evidence.
- Hence, Type I error is typically considered to have more serious consequences than Type II error.
- Jerzy Neyman proposed the **significance test** (or **test of significance**, 显著性检验) with the basic idea: try to minimize the Type II error rate provided that Type I error rate is below a given level.

Significance Test

- Let $\mathbf{X} = (X_1, \dots, X_n)$ be a simple random sample from the population $X \sim f(x; \theta)$.
- For testing problem $H_0: \theta \in \Theta_0 \leftrightarrow H_1: \theta \in \Theta_1$ and $\forall \alpha \in (0, 1)$, if a test has rejection region RR such that

$$P(\mathbf{X} \in RR) \leq \alpha, \forall \theta \in \Theta_0,$$

α is the maximum acceptable Type I error rate

- then the test is called a **significance test of θ with significance level α** (显著性水平为 α 的显著性检验), or simply **a test with level α** (水平为 α 的检验).

- Commonly used values of α are 0.1, 0.05, 0.01.



5.3 Hypothesis Testing

Solution of Example 5.8

- The testing problem is $H_0: \mu \leq 175 \leftrightarrow H_1: \mu > 175$. First, consider a simpler testing problem:

$$H_0^*: \mu = 175 \leftrightarrow H_1: \mu > 175.$$

- Based on our previous thinking, the rejection region is of the form $\{\mathbf{X}: \bar{X} - 175 > c\}$. The value of c need to satisfy that the Type I error rate is controlled at α , i.e.,

$$\alpha = P(\bar{X} - 175 > c | H_0^* \text{ is true}).$$

- To determine the value of c , we need to refer to the distribution of $\bar{X}$ when H_0^* is true, i.e., $\mu = 175$.
- By the CLT, we have the distribution of $\bar{X}$ under H_0^* :

$$\bar{X} \sim N\left(175, \frac{\sigma^2}{n}\right) \Rightarrow Z = \frac{\bar{X} - 175}{\sigma/\sqrt{n}} \sim N(0, 1).$$

- Then

$$\alpha = P(\bar{X} - 175 > c | H_0^* \text{ is true}) = P\left(Z > \frac{c}{\sigma/\sqrt{n}} \middle| \mu = 175\right) \Rightarrow c = z_\alpha \times \frac{\sigma}{\sqrt{n}}.$$



5.3 Hypothesis Testing

Solution of Example 5.8

- Next, consider the original null hypothesis $H_0: \mu \leq 175$, we find that for any $\mu \leq 175$, the test with $RR = \{\mathbf{X}: \bar{X} - 175 > z_\alpha \sigma / \sqrt{n}\}$ can control the Type I error rate under α :

$$P\left(\bar{X} - 175 > z_\alpha \times \frac{\sigma}{\sqrt{n}} \mid H_0 \text{ is true}\right) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > z_\alpha + \frac{175 - \mu}{\sigma/\sqrt{n}} \mid \mu \leq 175\right)$$

$$\stackrel{N(0,1)}{\leq} P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > z_\alpha \mid \mu \leq 175\right) = \alpha.$$

- Therefore, the test with $RR = \{\mathbf{X}: \bar{X} - 175 > z_\alpha \sigma / \sqrt{n}\}$ is also a significance test with level α for the testing problem $H_0: \mu \leq 175 \leftrightarrow H_1: \mu > 175$.
- Plugging in $\alpha = 0.05$, $\sigma = 5$, $n = 25$, the RR is $\{\mathbf{X}: \bar{X} - 175 > 1.645\} = \{\mathbf{X}: \bar{X} > 176.645\}$.
- Based on the observed sample, we have $\bar{x} = 177$, which falls into the RR.
- Therefore, we would like to reject H_0 at significance level 0.05, i.e., the mean height of all male students in SUSTech is significantly greater than 175cm at level $\alpha = 0.05$.



5.3 Hypothesis Testing

- Example 5.8 suggests that when there are multiple parameter values in H_0 (e.g., $H_0: \mu \leq 175$), it can be simplified to a **simple null hypothesis (简单原假设)** involving only one parameter value which is closest to H_1 (e.g., $H_0^*: \mu = 175$).
- This is why most hypothesis testing problems are formulated with a simple null hypothesis.
- Note that the decision made on whether to reject H_0 depends on the significance level α .
 - E.g., in Example 5.8, if $\alpha = 0.01$, then the RR is $\{X: \bar{X} > 177.33\}$, the decision is “fail to reject H_0 ”.
 - So, simply providing a conclusion on whether to reject H_0 may not be sufficient, we need to quantify our willingness to reject H_0 . The well-known **p-value (p值)** serves as such a quantitative measure.
- **p-value (p值)** is a key concept in hypothesis testing, from Wikipedia:

In statistical hypothesis testing, the **p-value** or **probability value** is the probability of obtaining test results at least as extreme as the results actually observed during the test, assuming the null hypothesis is correct.

- **$p\text{-value} = P(\text{observed or more extreme data} | H_0 \text{ is true})$.**



5.3 Hypothesis Testing

- The idea of p -values has been used in the story of *The Lady Tasting Tea*, we obtained that “under the assumption that the lady has no ability to distinguish between the two ways, the probability of her correctly identifying the 4 cups where tea was added first is only 0.0143.”

Assuming H_0 is true

Observed or more extreme data

The p -value

- A small p -value indicates that it is highly unlikely to observe the data we actually observed if H_0 is true. So, we tend to reject H_0 if p -value is small.
- The smaller the p -value, the stronger evidence we have against H_0 .
- Given a significance level α , we would reject H_0 if $p\text{-value} < \alpha$, otherwise, we fail to reject H_0 .
- A common misinterpretation of the p -value: $p\text{-value} = P(H_0 \text{ is true} \mid \text{observed data})$. ❌



5.3 Hypothesis Testing

Solution of Example 5.8

- Again, consider the simplified testing problem: $H_0^*: \mu = 175 \leftrightarrow H_1: \mu > 175$.

- The p -value is calculated as

$$P(\bar{X} \geq 177 | H_0^* \text{ is true}) = P\left(\frac{\bar{X} - 175}{\sigma/\sqrt{n}} > \frac{177 - 175}{\sigma/\sqrt{n}} \middle| \mu = 175\right) = P(Z > 2) = 0.023.$$

- Since the p -value = $0.023 < \alpha = 0.05$, we would reject H_0 at significance level $\alpha = 0.05$.
- On the other hand, if $\alpha = 0.01$, then p -value = $0.023 > \alpha$, we fail to reject H_0 at level $\alpha = 0.01$.



5.3 Hypothesis Testing

- Hypothesis testing is a powerful tool that provides a framework for assessing the statistical significance of hypotheses based on observational data.
- The application of hypothesis testing is very broad, encompassing fields such as scientific research, quality control, medicine and healthcare, economics and finance, psychology and social sciences, environmental studies, and market research.
- However, it is important to acknowledge that hypothesis testing have **inherent deficiencies**, and thus, a **critical mindset** should be maintained when interpreting the results of these tests.
- First, even we control the Type I error rate to be below α , it does not mean that the testing conclusion is 100% correct, as low-probability events can still occur.
- This is the reason why you may see contradictory conclusions in scientific research.
- Let's simulate the occurrence of Type I error in RStudio for a better understanding.



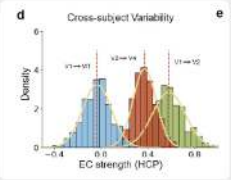
5.3 Hypothesis Testing

- Second, hypothesis testing can only assess **statistical significance** (统计显著性), but this does not imply **practical significance** (实际显著性).
 - When the sample size is large enough, even minor differences can be detected, but these differences may not be significant in practical terms.
 - Practical significance should be evaluated by domain experts relevant to the research question.



5.3 Hypothesis Testing

2月1日 23:08



徐老师，请教一个很傻的统计上的问题哈

我们有批数据算出来的大脑的不同脑区的连接

这个左边蓝色的，我感觉明显就跟0比做独立样本 ttest 会不显著，但是因为样本量大， $n=800$ ，学生算出来就是显著的，很显著

2月1日 23:11

我算了一下， $\text{mean}=-0.04$ ， $\text{std}=0.12$ ， $n=800$ ，也确实是显著的。

但是很不符合我对大脑的理解。蓝色的这个分布看着，不应该显著呀。

统计上的显著性跟实际意义上的显著性是不一样的

那怎么来理解，统计上显著，但是直觉上看着一点儿也不显著呢？

在样本量很大的情况下，细小的差距可能在统计意义上就是显著的

没事没事，谢谢徐老师回复！这么晚，打扰您了。

匆少侠也在江湖：不好意思，刚刚看到～

这个说得应该比较清楚～

匆少侠也在江湖：

收到

这个 practical significance 也是会算出来一个 p 吗？然后我们 report 这个 p value?

2月2日 00:17

不是哦，practical significance 更像是具体问题的专家给出的判断

这个 practical significance 也是会算出来一个 p 吗？...

2月2日 00:18

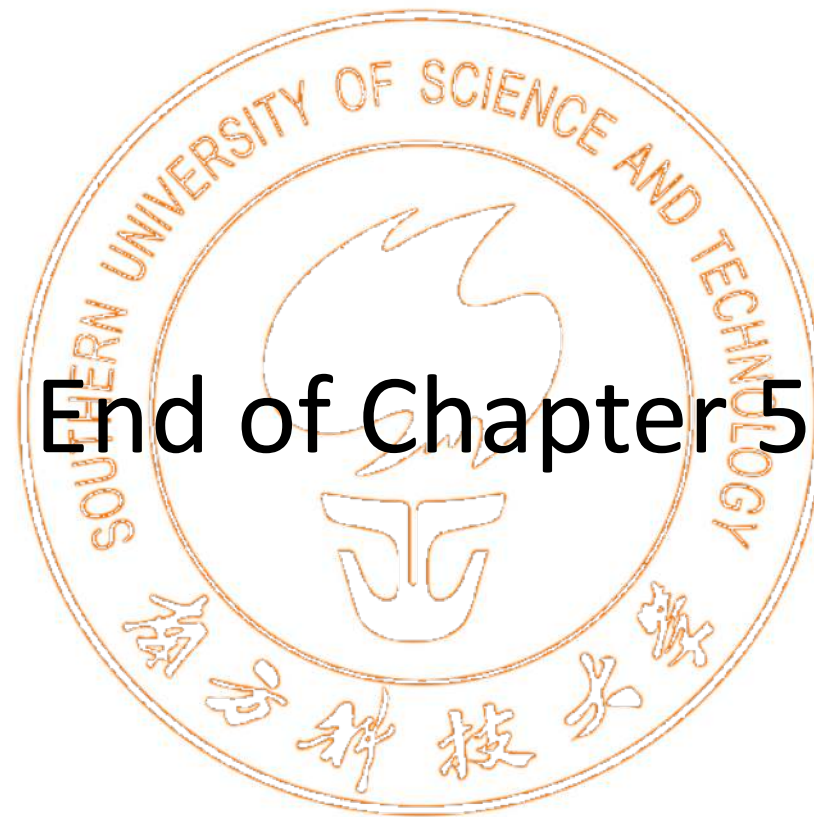
statistical significance 是从概率的角度认为观察到的差距不是由于随机抽样导致的，而是真实存在的差距

practical significance 是在判断这个差距在具体的场景下是不是有实际意义

哦哦哦

practical significance 也就是说，有没有实际意义，由专家说了算





End of Chapter 5